

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2024 Course: B. Tech. Semester: V Branch: Mechanical / Automobile / Production / Mechanical Engineering Sandwich Subject Code & Name: Automobile Engineering BTAPE504D Max Marks: 60 Date: 08/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	What are the main components of an automobile? Describe all of them briefly.	CO1	6
B)	How do you classify automobiles? Explain in detail.	CO2	6
C)	Explain briefly the various types of chassis construction with the help of suitable diagram.	CO3	6
Q.2	Solve Any Two of the following.		12
A)	Sketch and explain the layout of steering system	CO2	6
B)	State the advantages and disadvantages of independent suspension over rigid axle type suspension.	CO2	6
C)	Why springs are provided in automobile suspension? And explain briefly the Different Spring used for suspension.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Describe the working of a “single plate clutch” with a neat sketch and write its advantages and disadvantages.	CO1	6
B)	With the help of a neat sketch, explain the construction and operation of a constant mesh gearbox.	CO2	6
C)	Explain the construction and working of a differential with a neat sketch.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	Explain the construction and working of mechanical brakes with a neat	CO2	6

	sketch.		
B)	Explain hydraulic brake with a neat sketch and write its advantages.	CO2	6
C)	What is anti-lock braking system? Explain the need and functioning of ABS with Sketch.	CO2	6
Q. 5	Solve Any Two of the following.		12
A)	Explain in Brief Magneto-coil and solid stage ignition systems	CO4	6
B)	Explain Battery charging system in automobile.	CO4	6
C)	What is pollution due to vehicle emission explain in brief.	CO5	6
	*** End ***		